ML Assignment 2

Introduction:

Our dataset is about healthcare and some information about persons.

Our problem is to predict if this person has stroke or not based on properties of them.

1. Data Preprocessing

We begin by preparing the dataset for modeling:

- Load the dataset using **Pandas**.
- Display basic structure using .head(), .describe(), and .info().
- Check for **missing values** using .isnull().sum().
- Verify correct **data types**.
- Ensure no **categorical features** need encoding.

2. KNN Model

- We started with selection **all** features without **stroke** and select stroke as y (target)
- We Split x values as 60% train, 20% validation and 20% test using train_test_split() from sklearn.model_selection
- We scaled x values to be in same ranges using StandardScaler() from sklearn.preprocessing
- We fitted our model to every k from 1:20 and see best k based in validation accuracy score
- Plot every validation accuracy based on k
- **Best k is 6**
- We trained our model in k = 6 and this final model

3. Cross-Validation:

We used 5-fold to compare many validation sets accuracy to our model

Results:

Cross-validation average accuracy is very close to our mode which is good

Why k-fold?

Cross-validation is usually more robust because**:**

- It uses different training-validation splits multiple times.
- It gives a more reliable performance estimate than just a single validation split.
- It helps to avoid the risk that you got "lucky" or "unlucky" with one particular validation set.
- The Test Set is never touched during model selection — only for final evaluation

4. Confusion Matrix Analysis:

We used confusion matrix to represent all TP, TF, FP, and FN so we can notice patterns.

Results:

- model predict only one true positive and he predicted 46 false negative
- in report the recall of positive diagnosis of is 2% which is very bad
- The model very biased to negative diagnosis due to imbalance of data (95% is negative)

Imbalance of data caused the model to be very biased to negative.

Unfortunately, it is hard to slove in our sitiuation and using KNN.

5. Overfitting Discussion:

- Actually, our model doesn't overfit due to training, test, validation and avg cross- validation doesn't differ much.
- training acc ~ 95% test & validation & cross-validation average ~ 95.4% so our model doesn't have any sign of overfitting

But we made code to improve our model slightly using Feature Selection Technique.

Why Feature Selection?

- **because it reduce complexity of model as we have many columns and few rows (5000)**
- **make model learn instead of memorize**

Implementing Feature Selection Technique:

- We used random forest and correlation matrix to choose most important and related features
- We choose most proper 4 features
- Retrain model with new columns and used same approach in first model

Result:

- Model improved slightly due to feature selection

Conclusion:

Model is not good because of available date which caused biased, so it much better to use techniques instead of KNN and collect more data.